

Review

Advancements in Parkinson's Disease Diagnosis: A Comprehensive Survey on Biomarker Integration and Machine Learning

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Abstract: This comprehensive review explores the advancements in machine learning algorithms in the diagnosis of Parkinson's disease (PD) utilizing different biomarkers. It addresses the challenges in the assessment of PD for accurate diagnosis, treatment decisions, and patient care due to difficulties in early and differential diagnosis, subjective clinical assessments, symptom variability, limited objective biomarkers, comorbidity impacts, uneven access to specialized care, and gaps in clinical research. This review provides a detailed review of ongoing biomarker research, technological advancements for objective assessment, and enhanced healthcare infrastructure. It presents a comprehensive evaluation of the use of diverse biomarkers for diagnosing Parkinson's disease (PD) across various datasets, utilizing machine learning models. Recent research findings are summarized in tables, showcasing key methodologies such as data preprocessing, feature selection, and classification techniques. This review also explores the performance, benefits, and limitations of different diagnostic approaches, providing valuable insights into their effectiveness in PD diagnosis. Moreover, the review addresses the integration of multimodal biomarkers, combining data from different sources to enhance diagnostic accuracy, and disease monitoring. Challenges such as data heterogeneity, variability in symptom progression, and model generalizability are discussed alongside emerging trends and future directions in the field. Ultimately, the application of machine learning (ML) in leveraging diverse biomarkers offers promising avenues for advancing PD diagnosis, paving the way for personalized treatment strategies and improving patient outcomes.

Keywords: Parkinson's disease; multimodal biomarkers; machine learning; biomarker integration



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1. Introduction

Parkinson's disease (PD) is a neurodegenerative disorder characterized by progressive motor and non-motor symptoms. Early and accurate diagnosis is crucial for timely intervention and management. Advancements in machine learning (ML) have enabled the exploration of diverse biomarkers to enhance PD diagnosis. This review examines the application of ML techniques in utilizing biomarkers such as speech analysis, electroencephalography (EEG), gait patterns, and functional magnetic resonance imaging (fMRI) to aid in the early detection and classification of PD.

The diagnostic process typically starts with an extensive clinical assessment performed by a neurologist, encompassing detailed review of medical history, comprehensive neurological examination, and thorough evaluation of cognitive and motor functions. Diagnosing idiopathic PD can be relatively straightforward when patients present with typical symptoms such as asymmetric motor signs, a classic clinical history, and no atypical features [1]. However, in routine clinical practice, misdiagnoses are common, with PD frequently confused with other tremor disorders, such as essential tremors or forms of secondary Parkinsonism. Differentiating PD from atypical Parkinsonian disorders including multiple

system atrophy (MSA), progressive supranuclear palsy (PSP), and cortico-basal degeneration (CBD) poses particular challenges in early stages due to overlapping symptoms, and even seasoned neurologists may struggle with accurate diagnosis in these cases, as defining features often emerge only later in disease progression. PD primarily results from the degeneration of dopaminergic neurons in the substantia nigra, a critical region of the midbrain, due to the accumulation of alpha-synuclein in Lewy bodies and neurites. This neuronal loss precedes clinical symptoms and manifests in motor symptoms, including slowness of voluntary movement (bradykinesia), 4–6 Hz rest tremor, muscular rigidity, and postural instability as well as non-motor symptoms including loss of smell, sleep disturbances, cognitive alterations, and depression. However, the sequence in which these symptoms develop varies significantly between patients. Neurologists assess motor and non-motor impairments (such as daily activities, mentation, and behavior) through a series of mechanical tasks and evaluations. The scores from these clinical assessments, particularly the modified Unified Parkinson's Disease Rating Scale (UPDRS-III)—the most widely used PD rating scale in the U.S.—help estimate disease severity. Other scales, including the modified Hoehn and Yahr (H&Y) scale (0 to 5) and the Schwab and England activities of daily living scale (0 to 100%), also gauge patient disability, with higher scores indicating greater impairment. These scales provide critical insights into individual disease burden but may vary in precision. To improve diagnostic accuracy, updated diagnostic criteria have been developed, such as those proposed by the International Parkinson and Movement Disorder Society, which offer two levels of diagnostic certainty and introduce supportive and exclusionary features.

Although these criteria improve diagnostic accuracy, misclassification still occurs, especially in early disease stages, emphasizing the need for better diagnostic tools and biomarkers. Monitoring symptom progression is critical in refining the diagnosis. Speech analysis, gait assessment, and fMRI play pivotal roles in diagnosing and understanding neurodegenerative diseases. These tools offer unique insights into functional impairments and underlying brain changes associated with these conditions, providing valuable information beyond traditional diagnostic methods. Breakthroughs in AI and ML supply promising tools for medical providers to diagnose PD more efficiently and at earlier stages by analyzing data from speech patterns, movement videos, and brain imaging, detecting subtle patterns often missed in traditional assessments. These technologies have the potential to enable earlier detection, timely treatment, and a semi-automated diagnostic process, improving convenience for both patients and doctors. Machine learning models have demonstrated promise in the early detection of PD, often before clinical symptoms manifest. By analyzing diverse data types—including clinical records, neuroimaging, and genetics—these methodologies can identify patterns and biomarkers associated with the disease. Early detection is crucial for initiating treatments that can potentially slow disease progression. Current research explores the application of machine learning to evaluate biomarkers such as speech and gait as potential diagnostic tools for PD. Numerous studies have focused on identifying optimal features from these biosignals linked to PD and other neurodegenerative diseases, employing various signal processing and machine learning methods from preprocessing and feature extraction to classification.

Motivation

The assessment of PD presents several challenges, which can impact diagnosis, treatment decisions, and patient management. Some of the current challenges include challenges in early diagnosis and differential diagnosis, subjectivity in clinical assessment, variability in symptoms and progression, and limited biomarkers for objective assessment.

- **Early Diagnosis and Differential Diagnosis:** Diagnosing PD can be difficult, especially in early stages when symptoms are subtle and overlap with other movement disorders. Differentiation from conditions like essential tremor or atypical Parkinsonian syndromes (e.g., multiple system atrophy, progressive supranuclear palsy) requires thorough clinical evaluation, and sometimes advanced imaging or biomarker tests.

- **Subjectivity in Clinical Assessment:** Traditional assessment of PD relies heavily on clinical judgment and subjective rating scales (e.g., Unified Parkinson's Disease Rating Scale, Hoehn and Yahr scale). This subjectivity can lead to variability in diagnosis and monitoring of disease progression.
- **Variability in Symptoms and Progression:** PD is a heterogeneous disease with a wide range of motor and non-motor symptoms, which can vary significantly between individuals. Symptoms may also fluctuate over time, complicating assessment, and treatment planning.
- **Limited Biomarkers for Objective Assessment:** While the research is advancing in biomarkers such as imaging techniques (e.g., fMRI, PET scans) and biochemical markers (e.g., alpha-synuclein levels in cerebrospinal fluid), reliable biomarkers for early and accurate diagnosis, as well as for tracking disease progression, are still limited.
- **Impact of Comorbidities:** Many PD patients have comorbid conditions (e.g., depression, cognitive impairment) that can complicate assessment and management. Understanding how these conditions interact with PD symptoms is crucial for comprehensive patient care.
- **Access to Specialized Care and Resources:** Access to neurologists specializing in movement disorders and comprehensive care facilities can vary geographically, leading to disparities in assessment quality and timeliness of interventions.
- **Clinical Trials and Research Gaps:** There remains a need for large-scale longitudinal studies and clinical trials to better understand PD progression and identify effective biomarkers and therapeutic strategies. Recruitment challenges and heterogeneity in study populations can also impact the generalizability of research findings.

Addressing these challenges requires continued research into biomarkers, advancements in technology for objective assessment, and improvements in healthcare infrastructure to support timely and accurate diagnosis and management of Parkinson's disease.

2. Literature Review

Numerous studies have contributed comprehensive overviews of AI-driven approaches aimed at enhancing diagnostic accuracy, exploring multiple biomarkers, and handling complex multimodal datasets. Early surveys in this area primarily focused on ML techniques for speech and motor analysis, as these were among the initial biomarkers studied for PD-related symptoms. Researchers have systematically reviewed various data types, such as acoustic, motion, and neuroimaging, emphasizing the value of non-invasive data collection and feature extraction as primary steps for identifying PD-related patterns. These initial surveys have helped shape the understanding of PD biomarkers and underscored the critical challenges, including data quality, patient variability, and the need for high-dimensional data handling. More contemporary reviews have taken a broader perspective by investigating the advancements in deep learning techniques that enable more sophisticated analysis of complex data types. In 2022, Haq et al. conducted a comprehensive evaluation of deep learning techniques for diagnosing PD using clinical data, detailing processes such as data preprocessing, feature extraction, and classification [2]. This study provides an analysis of numerous ML models, provides model performance metrics and validation techniques, addresses data variability and feature selection challenges, and advocates for integrating these methods with E-health systems to improve accurate diagnosis. An extensive 2023 review explores recent advances in using deep learning for PD diagnosis, focusing on EEG, MRI, speech, handwriting, and sensor data from 2016 to 2022 [3]. Highlighting models like CNNs and RNNs, it shows their potential in identifying biomarkers and improving disease staging particularly for analyzing complex datasets such as time-series EEG data. The paper also addresses current challenges, such as data quality and model generalization, and suggests directions to strengthen deep learning's role in PD diagnosis. The review by Dixit et al. [4] surveys machine learning and deep learning models for diagnosing PD, evaluating AI approaches across multimodal datasets, including speech, motor symptoms, and handwriting on different types of patient data, such as imaging and wearable sensor

data. It highlights challenges like data scarcity and model explainability, suggesting that integrating omics and health record data could enhance early detection and diagnostic accuracy. In their systematic review of 50 research articles focused on PD diagnosis, Pradeep et al. analyze various diagnostic modalities such as imaging, signal processing, and data analysis, emphasizing the role of machine learning and deep learning approaches. Their paper highlights existing datasets, tools, and performance measures used in diagnosing PD while identifying research gaps and challenges in current methodologies [5]. They also analyze specific performance metrics, evaluate the highest-performing methods, and provide a chronological review of the selected studies.

This review highlights advancements in machine learning for PD diagnosis across diverse biomarkers, with a particular emphasis on the integration of ML techniques for early detection through a multi-biomarker approach. This focus offers a potential in advancement over traditional diagnostic methods, which often depend on subjective clinical evaluations. In this section, we discuss the role of speech analysis as a non-invasive biomarker, emphasizing the extraction of vocal features indicative of PD-related dysfunctions with the implementation of various machine learning algorithms. Next, EEG-based studies are explored, focusing on spectral analysis and machine learning algorithms to detect abnormal brain activity patterns associated with PD. Additionally, fMRI studies are reviewed for their ability to capture functional brain alterations in PD patients, highlighting ML's role in analyzing complex neuroimaging data to uncover disease-related patterns. Gait analysis, another widely investigated biomarker, is explored and the studies leveraging ML to identify characteristic gait changes and distinguish PD from healthy controls or other movement disorders are reviewed. In Section 3, we will examine the limitations associated with diagnostic paradigms that rely on individual biomarkers, as well as the shortcomings of existing trends and datasets. Following this, in Section 4, we will delve into current efforts aimed at integrating biomarkers in the diagnosis of PD, also exploring various integration techniques that can be implemented for the early detection of PD.

2.1. Speech in PD Diagnosis with the Application of Machine Learning

Early efforts to utilize speech processing for Parkinson's disease (PD) classification initially involved manually analyzing acoustic features of speech signals. During the 1990s, researchers employed acoustic analysis techniques to distinguish speech from PD patients and healthy controls based on characteristics such as pitch, duration, and intensity. The advancement of machine learning and artificial intelligence has significantly automated and enhanced the accuracy of speech processing in PD classification. By the mid-2000s, researchers had begun integrating machine learning algorithms to automatically classify speech signals as belonging to PD patients or healthy controls.

In 2011, Little et al. [6,7] introduced dysphonia measures, a novel speech signal processing algorithm aimed at predicting PD symptom severity using speech signals. Their study evaluated the effectiveness of these measures in discriminating PD subjects from healthy controls using four feature selection algorithms and two statistical classifiers. The findings indicated that some of these dysphonia measures complement existing algorithms, thereby enhancing classifiers' capability to distinguish between healthy controls and PD subjects. The feature sets from these early studies laid the foundational groundwork for subsequent developments and some of the most significant features are listed in Table 1. A novel approach to feature extraction has been proposed by [8], focusing on the pitch-synchronous level rather than the conventional block segmentation with fixed frame lengths. In a modified study [9], an unsupervised k-means clustering algorithm was employed for classification purposes. With the novel feature extraction technique, a set of temporal, and spectral features were proven to perform superior for PD classification after testing with 17 different ML classifiers, even in continuous speech data [10]. The authors utilize two datasets: one dataset containing 40 participants, 22 (7 female and 15 male) PD and 18 (12 female and 6 male) healthy controls (HCs), and the other one being the Italian Parkinson's Voice and speech dataset, which is available in IEEE DataPort [11]. The imbalance of

both datasets in terms of test and control groups, as well as their variation in languages, demonstrates the robustness of the proposed methodologies and selected features.

Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have emerged as pivotal tools in PD detection and classification using speech signals. These models excel in automatically extracting pertinent features from speech signals and utilizing them to classify based on inherent data patterns. For instance, one method involved analyzing features extracted from sustained phonation recordings captured with an acoustic cardioid (AC) and a smartphone (SP), employing MLP, KNN, and SVM models. This approach achieved accuracies of 94.55% and 92.94%, with corresponding AUC values of 87.84% and 92.40% [12]. The dataset for this study, collected by Sarkar et al. [13] at Istanbul University, consists of a training database with data from 20 patients with Parkinson's disease and 20 healthy subjects, alongside a separate testing database. In another study, support vector machine (SVM) with C-SVC and radial basis kernel functions successfully classified various stages of PD and healthy control (HC) based on the continuous speech recordings of 43 patients across various UPDRS grades (1 to 4) and 9 control subjects, all recorded reading the "Rainbow Passage", achieving an impressive overall average accuracy of 81.8% [14]. Further advancements included training and evaluating SqueezeNet1_1, ResNet101, and DenseNet161 architectures for PD detection, where DenseNet-161 attained the highest accuracy of 89.75%, sensitivity of 91.50%, and precision of 88.40% [15]. This study's PD dataset was sourced from the mPower public data portal, where participants provided consent via the mPower app (launched in 2015) to assess PD symptom severity and medication sensitivity. A research study in 2021 proposed using a Bidirectional Long Short-Term Memory (LSTM) model to capture dynamic time-series features by analyzing energy transitions during shifts from unvoiced to voiced speech segments custom dataset comprising 45 subjects (15 HCs and 30 PD patients with H&Y stages 1–5), collected from volunteers at the GYENNO SCIENCE Parkinson Disease Research Center [16]. Another study integrated voice features into distinct frameworks, emphasizing feature integration at various levels, validated through Leave-One-Person-Out Cross-Validation (LOPO CV) on a dataset accessed from UCI machine learning repository [17]. Additionally, a novel LDA-NN-GA framework combining LDA and Genetic Algorithm (GA) achieved notable accuracies of 95% in training and 100% in testing on sustained phonation datasets [18]. Moreover, Chen et al. [19] applied feature reduction techniques using PCA and Fuzzy KNN to enhance the precision of PD voice signal detection to approximately 96.07% using datasets from the UCI machine learning repository. The Deep Multi-Variate Vocal Data Analysis (DMVDA) System, incorporating Acoustic Deep Neural Network (ADNN), Acoustic Deep Recurrent Neural Network (ADRNN), and Acoustic Deep Convolutional Neural Network (ADCNN), improved the analysis of speech attributes, achieving an impressive classification accuracy of 98% [20]. These advancements summarized in Table 2 highlight the significant strides made in leveraging advanced computational techniques to enhance PD diagnosis and monitoring through speech analysis.

The following bullet points summarize advancements in using ML for PD diagnosis through speech analysis.

- Early speech processing methods relied on manually analyzing features like pitch, duration, and intensity to distinguish PD patients from healthy individuals, establishing foundational dysphonia measures later used for PD severity estimation and feature selection with statistical classifiers.
- Pitch-synchronous feature extraction emerged as an innovative approach, showing strong classification performance on continuous speech data, even with imbalanced datasets.
- CNNs and RNNs, particularly DenseNet-161 and LSTM-based models, now play a key role in PD detection from speech, achieving high accuracy and sensitivity.
- Various methods like SVM, MLP, KNN, and ensemble models have shown diverse accuracies (e.g., SVM with radial basis kernel reached 81.8% AUC; DenseNet achieved

~90% accuracy), highlighting the impact of different feature extraction and modeling techniques.

- Advanced frameworks, including LDA-NN-GA, PCA with Fuzzy KNN, and multi-variate deep networks (ADNN, ADRNN, ADCNN), have further pushed classification accuracy, with recent deep learning frameworks reaching up to 98% accuracy on certain datasets.

Table 1. Key speech and EEG features for PD identification in traditional and machine learning approaches.

Biosignal	Parameters Significant for PD Identification
Speech	Pitch period [6–10], LOC (length of curve) [8,10], peak frequency [6–10], energy [2–6], MFCCs [6–10], correlation canceler efficiency [8,10], jitter [6,7], shimmer [6,7].
EEG	Alpha band power [21], alpha asymmetry [21], beta band power [21], delta and theta band power [22], gamma band oscillations [23], connectivity coherence [23], PAC patterns [23], Approximate Entropy (ApEn) [24], Hurst coefficient [25], higher-order statistics [25–28].

Table 2. A summary of the state-of-the-art machine and deep learning methods applied on speech dataset for PD diagnosis.

Method	Model	Performance	Reference
Feature selection, statistical classifier	Speech signal	Clinically useful accuracy (about 7.5 UPDRS points difference from the clinicians' estimates)	[6,7]
Coarse Gaussian SVM, ensemble boosted tree, logistic regression	Speech features (Pitch synchronous and MFCC) for PD relevant feature identification	Identification of optimal feature set. Median accuracies up to 94%	[8–10]
MLP, KNN, SVM	Feature extracted from sustained phonation	94.55%, 92.54% (AUC: 87.84%, 92.4%)	[12]
SVM with C-SVC, radial basis kernel	Continuous speech recording	81.8% AUC	[14]
SqueezeNet1_1, ResNet101, DenseNet161		Accuracy 89.75%, sensitivity 91.50%, precision 88.40%	[15]
Bidirectional LSTM	Dynamic time-series features		[16]
LDA-NN-GA		Accuracy 95–100%	[18]
PCA, Fuzzy KNN	Voice signal reduced features	96.07% accuracies	[19]
ADNN, ADRNN, ADCNN	Speech attributes	98% accuracy	[20]

These advancements demonstrate the potential of ML-based speech analysis as a non-invasive, accurate tool for PD detection and monitoring in clinical and remote settings.

2.2. EEG in PD Diagnosis with the Application of Machine Learning

Identifying Parkinson's disease (PD) via EEG involves extracting specific neural features associated with the disease. While EEG alone may not diagnose PD definitively, the EEG along with the relevant features as listed in Table 1 in Section 2.1. can provide valuable insights when combined with clinical assessments and other biomarkers. Key EEG patterns include decreased alpha band power in posterior brain regions, abnormal frontal alpha asymmetry linked to PD-related depression, increased beta band power [21] in the motor cortex related to motor symptoms, and elevated delta and theta band power in frontal regions associated with cognitive deficits [22]. Other indicators like gamma-band oscillations, ERP P300 fluctuations, reduced connectivity coherence, abnormal PAC patterns [23], decreased EEG signal complexity (e.g., ApEn) [24], changes in spectral en-

tropy, fractal dimension, and SL for altered connectivity further aid in understanding PD neurophysiology.

Current trends in machine learning for PD diagnosis via EEG focus on leveraging advanced neural network architectures tailored for EEG data to learn complex patterns, integrating EEG with other biomarkers to enhance diagnostic robustness, adopting real-time monitoring with wearable EEG devices for early symptom detection and personalized treatment, exploring data augmentation and transfer learning to improve model performance with limited EEG datasets, enhancing model interpretability for clinicians, and emphasizing rigorous validation for clinical adoption, aiming to improve PD diagnosis accuracy and treatment outcomes. These approaches generally utilize automated feature extraction, a variety of machine learning algorithms, and neural network architectures like CNNs and RNNs to achieve precise PD detection, monitoring progression, and defining treatment paradigms. Wagh et al. [29] introduced an eight-layer graph convolutional neural network (CNN) that was trained on EEG feature matrices, leading to the attainment of an 85% AUC for the detection of neurological diseases, including PD. The study uses two large publicly available scalp EEG databases: the TUH EEG Corpus [30], containing clinical recordings of patients with neurological disorders, and the MPI LEMON Dataset [31], comprising resting-state recordings from healthy participants. In 2018, three models were proposed by Vanegas et al. [32] to identify EEG biomarkers for PD. The initial model, featuring an extra tree classifier, demonstrated remarkable performance with a 99.4% AUC in distinguishing PD subjects from control subjects based on EEG spectral amplitudes. In another study [33], a random forest (RF) model is used to categorize PD achieving a remarkable 91% AUC. This study's data includes EEG recordings from 125 patients, with 40 patients selected for group classification based on 'good' or 'poor' cognition, where five EEG segments per patient were extracted at a 500 Hz sampling rate. Hybrid CNN-RNN models for PD detection are proposed in [34,35], with 82.89% and 96.9% accuracy, respectively. The hybrid models demonstrated superior performance compared to the traditional ones. Specifically, 3D-CNN-RNN achieved the highest five-fold average accuracy at 82.89%, followed by 2D-CNN-RNN at 81.13%, CNN at 80.89%, and RNN at 76.00%. A 13-layer CNN was introduced by the authors for the classification of PD using resting-state EEG data, resulting in an accuracy of 88.3%, a sensitivity of 84.7%, and a specificity of 92% [36]. Various methods based on the automated tunable Q wavelet transform technique were employed by Khare et al. [37], including LSSVM, to differentiate between HC individuals and PD patients, both with and without medication. Remarkably, 96% and 97.7% accuracy rates were achieved for these distinctions on an open-source dataset of openneuro collected at the University of San Diego, California. A similar study was performed in [38] using common spatial patterns and entropy from EEG with various classifiers namely random forest, discriminant analysis, SVM, and KNN, where the accuracies ranged from 95% to 98% with KNN achieving the highest accuracies across all cases. In a later study by Khare et al. [39], with the assistance of smoothing of EEG using smoothed pseudo-Wigner Ville distribution (SPWVD) coupled with convolutional neural networks (CNN), accuracies of 99.97% and 100% for two different datasets were attained. Shaban et al. developed a deep learning framework with 98% accuracy, 97% sensitivity, and specificity of 100%. Additionally, they achieved up to 99.9% accuracy using a Wavelet-based CNN on openneuro EEG data [40,41]. In a separate investigation, 2D-CNN was utilized on EEG data transformed using the Gabor technique, resulting in a remarkable accuracy of 99.5% for the PD classification task [42,43]. They have also utilized the EEG data available on openneuro for their studies.

EEG event-related potential ERP is also used in a study [25], for early PD detection with a novel Brain Network Analytics (BNA) technique and logistic regression for classification with a false positive rate (FPR) feature selection method during each iteration of classifier training. Various ML models, including decision tree (DT), discriminant analysis (DA), logistic regression classifier, support vector machine (SVM), k nearest neighbor (KNN), ensemble boosted trees, ensemble bagged trees, and ensemble subspace-KNN, were compared based on higher-order statistic (HOS) features extracted from EEG data.

The evaluation, employing AUC and ROC metrics, confirmed the superior performance of HOS features [26–28]. It is crucial to emphasize that EEG-based PD detection should be viewed as a complementary tool to clinical assessments and other diagnostic techniques, with the availability of diverse datasets being essential for training robust models. Table 3 presents the methods, models used, performance metrics, and references for studies focusing on EEG-based Parkinson’s disease detection using machine learning and deep learning techniques.

Table 3. A summary of the state-of-the-art machine and deep learning methods applied on EEG dataset for PD diagnosis.

Method	Model	Performance	Reference
8-layer graph convolutional neural network (CNN)	EEG feature matrices	85% AUC	[29]
Extra tree classifier	Distinguishing PD from control subjects based on EEG spectral amplitudes	99.4% AUC	[32]
Random forest (RF)	PD categorization	91% AUC	[33]
Hybrid CNN–RNN models	PD detection	Accuracies 82.89%, 96.9%	[34,35]
13-layer CNN	Resting state EEG for PD screening	88.3% accuracy, 84.7% sensitivity, 92% specificity	[36]
LSSVM and Q-wavelet-based automated method	PD classification	96%, 97.7% accuracy	[37]
Common spatial pattern and entropy with classifiers	PD classification	95–98% accuracies	[38]
SPWVD with CNN (PDCNet)	Smoothing of EEG relevant for PD	99.97%, 100% accuracies	[39]
Wavelet-based CNN	EEG smoothing and PD classification	98% accuracy, 97% sensitivity, and up to 99.9% accuracy respectively	[40,41]
2D-CNN on Gabor-transformed EEG data (Gabor PD Net)	PD classification	99.5% accuracy	[42,43]
ERP with Logistic regression, Brain Network Analytics (BNA) technique	FPR feature selection method Early-stage detection	AUC ROC of 0.79	[25]
Various ML models (DT, DA, logistic regression, SVM, KNN, ensemble methods)	Higher-order statistics (HOS) features extracted from EEG	AUC ROC 0.75, 0.8, and 0.89 respectively	[26–28]

In summary, EEG analysis, combined with ML, enhances the diagnosis of PD by detecting neural patterns associated with the disease. The key points from the above table are summarized in the bullet points below, which also describe developments in ML for PD diagnosis using EEG.

- Key EEG biomarkers for PD include decreased alpha power, increased beta power in the motor cortex, elevated delta/theta power in frontal regions, gamma oscillations, and altered signal complexity and connectivity.
- Advanced ML models, such as tree-based classifiers and neural networks (CNNs, RNNs), have been used for EEG-based PD detection. Vanegas et al. achieved 99.4% AUC using an extra tree classifier on spectral amplitudes.
- CNNs and hybrid CNN-RNN models are notable for high accuracy, with some hybrid models reaching 96.9% accuracy, and a 13-layer CNN achieving 88.3% for resting-state EEG data.
- Techniques like Q-wavelet transforms and the Gabor method, combined with CNNs, demonstrated strong performances, with accuracy rates up to 99.5–100%.

- Studies often use large EEG datasets like the TUH EEG Corpus, MPI LEMON Dataset, and openneuro data. These datasets support model robustness by providing diverse EEG recordings from both PD patients and healthy controls.
- ERP metrics and BNA techniques with classifiers like logistic regression aid early PD detection, with some models reaching AUCs up to 0.79.
- Various classifiers (e.g., DT, DA, SVM, KNN) on HOS features from EEG show AUCs from 0.75 to 0.89, highlighting EEG's role in identifying PD-related neural patterns.

2.3. fMRI in PD Diagnosis with the Application of Machine Learning

Functional magnetic resonance imaging (fMRI) has become a critical tool in the study of Parkinson's disease (PD), offering profound insights into the neural underpinnings of this progressive neurodegenerative disorder. fMRI operates by detecting changes in brain activity through monitoring variations in cerebral blood flow, leveraging the intrinsic relationship between neuronal activation and hemodynamic responses. When specific regions of the brain are engaged, a corresponding increase in blood flow occurs, facilitating the mapping of functional areas of the brain. This technique has proven particularly valuable in the early diagnosis of PD by detecting abnormal patterns of brain activity. Furthermore, fMRI plays a crucial role in distinguishing PD from other movement disorders that present with overlapping clinical features, such as essential tremor and atypical Parkinsonian syndromes [44–46]. One of the key contributions of fMRI is its utility in examining functional connectivity across brain networks. It enables the identification of disruptions in neural pathways specific to PD, offering deeper insights into how the disease alters the functional architecture of the brain [47,48]. Studies utilizing fMRI during motor task performance have significantly advanced our understanding of the neural circuits that regulate motor control, shedding light on the neural mechanisms behind hallmark PD symptoms such as bradykinesia, tremors, and rigidity. In addition to motor symptoms, cognitive decline, which is common in the later stages of PD, can also be investigated using fMRI. It provides critical data on alterations in brain activity associated with executive dysfunction, memory impairments, and other cognitive deficits. Additionally, fMRI has been instrumental in evaluating the effects of pharmacological treatments and other therapeutic interventions on brain function in individuals with PD, offering objective measures of therapeutic efficacy [49].

Machine learning and deep learning techniques have gained considerable momentum in neuroimaging studies of neurological and psychiatric disorders, including PD. These approaches have shown varying levels of success. Classical ML algorithms, such as support vector machine (SVM), have demonstrated promising results in PD research [50]. However, SVM's performance is often constrained by the need for manual feature extraction, particularly when applied to raw neuroimaging data. This limitation has catalyzed the growing integration of deep learning (DL) models, which offer automated feature extraction and functional connectivity analysis, making them highly effective for PD classification based on fMRI data. For instance, researchers have employed binary support vector machine (bSVM) and multiple-kernel learning (MKL) on diffusion tensor imaging (DTI) MRI data to classify PD patients. In one study, a dataset comprising 162 PD patients and 57 healthy controls was analyzed to select specific diffusion metrics, though the resulting area under the curve (AUC) of less than 60% indicated that DTI data alone may not be sufficient for accurate PD classification [51]. This study analyzed DTI data from the open-access PPMI database, including 162 patients with PD and 70 age- and gender-matched healthy controls, with PD patients averaging 63.9 years, 6.5 months disease duration, and a mean Hoehn and Yahr stage of 1.2. Innovative methodologies have also emerged, such as deep neural networks with broad views (DBV), which incorporate two aggregated residual transformation (ResNeXt) networks to analyze de novo PD patients from two perspectives: axial (AXI) and sagittal (SAG). This model achieved an accuracy of 76.46% on the PPMI dataset in screening PD patients, with the use of Wasserstein generative adversarial networks (WGANs) for data augmentation further improving the model's performance [52]. Another study [53] tested

various autoencoder architectures on clinical features obtained from the PPMI dataset (129 PD patients and 57 HCs from five clinical centers), mean diffusivity, and fractional anisotropy, with the spatial autoencoder attaining the highest AUC of 83%, highlighting the potential of autoencoders in capturing PD-specific brain patterns. Kazeminejad et al. [47] employed machine learning techniques to generate brain network graphs using diverse thresholding methods. By applying a feature selection approach, they identified the five most relevant metrics linked to a specific brain region, developing a classifier that achieved approximately 95% accuracy during leave-one-out cross-validation (LOOCV).

An automated classifier was also developed to differentiate between PD motor subtypes [54]. This study involved 96 PD patients divided into training and validation sets, with the most effective SVM model leveraging multilevel resting-state fMRI (rs-fMRI) indices. Functional activity and connectivity within the frontal lobe and cerebellum were identified as critical markers for distinguishing between PD subtypes. In another noteworthy investigation, a boosted logistic regression model was employed to distinguish PD patients from healthy controls using rs-fMRI data. The model utilized correlation matrices derived from fMRI data and incorporated a nested cross-validation approach with 10 outer and 10 inner folds, ensuring unbiased performance evaluation [55]. Data-driven methods for estimating functional networks, such as spatial independent component analysis (ICA) and principal component analysis (PCA), have also gained popularity, as they do not require predefined brain regions or voxels. Vieira et al. [56] utilized multilayer perceptrons (MLPs) to explore the relationship between psychiatric and neurological disorders, while Guçlu et al. [57] used recurrent neural networks (RNNs) to model the dynamics of human brain activity in response to sensory stimuli on two different datasets generated by Nishimoto et al. [58]. These RNN models processed feature sequences through two recurrent nonlinear layers and one linear layer for low-level visual feature encoding, with Long Short-Term Memory (LSTM) and gated recurrent units (GRUs) used for encoding high-level semantic information. Despite their power, RNN architectures typically require large datasets due to their substantial number of free parameters, with techniques like dropout often employed to mitigate overfitting. For example, Lebo Wang et al. [59] demonstrated that convolutional RNN (conv-RNN) achieved higher accuracy (98.5%) in individual identification using rs-fMRI compared to traditional RNN models (94.3%). The resting-state fMRI data for 100 subjects from the Human Connectome Project (HCP) [60] was used in this work. Conv-RNN excels in integrating both spatial and temporal features, allowing for more precise extraction of local features between neighboring regions of interest (ROIs). Unlike conventional RNNs, conv-RNN uses convolution operations in both the input-to-state and state-to-state transitions, replacing the Hadamard product. This design enables conv-RNN to capture temporal information linked to evolving features in its hidden states, providing deeper insights into functional interactions within the brain. In another study, Han et al. [61] applied a variational autoencoder (VAE) with a five-layer encoder and decoder to extract visual representations from a diverse set of unlabeled images. Using the VAE, they predicted and decoded cortical activity recorded via fMRI while subjects passively viewed natural videos. While the VAE achieved similar accuracy to convolutional neural networks (CNNs) in predicting cortical responses in early visual areas, its accuracy was lower in higher-order visual regions. Unsupervised techniques like ridge regression have also been employed for multivariate linking, where features from fMRI images are computed and classified using fully connected layers of pre-trained CNNs while subjects view dynamic or static stimuli [62]. Riaz et al. introduced FCNet, a novel network designed to compute functional connectivity from fMRI time series signals [63]. FCNet comprises a feature extractor with convolutional layers and a similarity measure network analogous to a Siamese network, followed by a softmax classifier. The study utilized resting-state fMRI data from the ADHD-200 consortium, comprising imaging and phenotypic data from three sites—NeuroImage (NI), New York University Medical Center (NYU), and Peking University—each with varying subject counts, and included separate training and independent testing datasets for each site [64]. Table 4 provides a comprehensive summary

of various methods, models, performance metrics, and references for studies focusing on neurodegenerative disease detection using machine learning and deep learning approaches with fMRI and other neuroimaging modalities.

Table 4. A summary of the state-of-the-art machine and deep learning methods applied on fMRI dataset for PD diagnosis and staging.

Method	Model	Performance	Reference
Brain network graphs	(i) Various thresholding methods within a machine learning framework (ii) Feature selection pinpointing relevant metrics linked to brain regions	95% accuracy	[47]
Conventional ML algorithm SVM	Expert-designed feature monitoring, poor performance on raw data	AUC ROC~99%	[50]
Binary support vector machine (bSVM) and multiple-kernel learning (MKL)	DTI MRI data for PD diagnosis	AUC < 60%	[51]
Deep neural networks with broad views (DBV): two aggregated residual transformation ResNeXt networks	PD screening	76.46%	[52]
Various auto-encoders	Clinical features, mean diffusivity, fractional anisotropy	AUC of ROC 83%	[53]
Automated classifier for motor subtypes of PD based on SVM with LSSVM	Multilevel rs-fMRI indices for PD motor subtype classification	AUC 0.917 on validation data	[54]
Boosted logistic regression model	rs-fMRI for PD screening	Mean accuracy 76.2%	[55]
MLPs and RNN	Utilizing fMRI for PD detection	83%	[56]
RNNs, LSTM, gated recurrent units with ridge regression	RNN with alternative feature model	94.3%	[57]
Conv-RNN	Integrating spatial and temporal features, extracting local features between neighboring ROIs	98.5%	[59]
Variational auto-encoder (VAE)	Extracting visual representations, predicting, and decoding cortical responses		[61]
FCNet	Brain functional connectivity calculation		[63]

The bullet points below summarize advancements in applying machine learning to PD diagnosis using fMRI, along with key insights from the table above.

- fMRI is instrumental in investigating PD, providing insights into abnormal patterns of brain activity, disruptions in functional connectivity, and distinctions between PD and other movement disorders.
- Classical ML models, such as SVMs, have been applied to PD diagnosis; however, their performance is often constrained by the requirement for manual feature extraction. More advanced models, including binary SVM and multiple-kernel learning (MKL) on DTI MRI data, demonstrated limited effectiveness, with an AUC of less than 60%.
- Deep learning models, particularly those that automate feature extraction, exhibit potential for PD classification. For example, deep neural networks with broad views (DBV) achieved an accuracy of 76.46% in screening for PD.
- Autoencoders, applied to clinical features and fMRI-derived metrics such as mean diffusivity, reached an AUC of 83%, underscoring their capability to capture PD-specific brain patterns.

- Automated classifiers leveraging resting-state fMRI data effectively identified PD motor subtypes, with SVM models achieving an AUC of 0.917, highlighting the relevance of rs-fMRI for analyzing motor symptoms in PD. A boosted logistic regression model attained a mean accuracy of 76.2% on rs-fMRI data for PD screening, demonstrating robust performance validated through cross-validation.
- RNNs and Conv-RNNs excel at modeling temporal and spatial features from fMRI data. Notably, Conv-RNNs achieved a high accuracy of 98.5% by capturing local features between neighboring ROIs.
- VAEs were used to extract visual representations and decode cortical responses, achieving moderate accuracy, thereby supporting the application of fMRI-based modeling for complex brain responses.
- FCNet utilizes fMRI time series signals to compute functional connectivity, contributing to an enhanced understanding of PD-specific brain network interactions.

2.4. Gait in PD Diagnosis with the Application of Machine Learning

Gait analysis provides critical insights into Parkinson's disease (PD) classification by examining temporal parameters (e.g., stride time, cadence, speed, stance duration, swing duration, and the swing/stance ratio) and spatial parameters (e.g., step length) alongside kinetic features such as maximum vertical forces at the initial contact and foot-off points, as outlined in Table 4. Machine learning (ML) has enabled the automated extraction and selection of these features for the early detection of PD [65]. Tremor, another prominent biomarker for PD, can be combined with gait analysis [66] to enhance the identification and diagnosis of PD. This is often achieved by comparing ground reaction force (GRF) plots between PD and healthy control (HC) subjects. Tremors associated with PD typically exhibit high amplitude and power concentrations in the 4–6 Hz range, which distinguish them from atypical tremors. Common ML algorithms used for PD classification via gait analysis include SVMs, random forest, and neural network [67–69], with linear discriminant analysis (LDA) also employed. Recursive feature elimination (RFE) with random forest and SVM is utilized to select clinically relevant gait features for PD classification [69]. The dataset for this experiment includes gait data collected using GYENNO Technologies' wearable sensor system from 200 PD patients and 100 HCs, with participants performing a 6 m "Timed Up and Go" (TUG) test while 10 sensors attached to various body parts record movement data, later processed into 95 gait characteristics. ANOVA tests are instrumental in distinguishing subjects based on the mean values of gait features, with significant differences observed between HC and PD subjects in metrics such as average step distance, stance phase, swing phase, heel force, and normalized heel force (with a 95% confidence interval and p -value ≤ 0.5 considered statistically significant) [67]. This study was validated on the Physionet dataset [70]. Cross-validation techniques, such as leave-one-subject-out (LOSO) validation, are essential for evaluating model performance and ensuring robustness when applied to new data.

For detecting Freezing of Gait (FoG)—a common and debilitating PD symptom where individuals experience sudden immobility a multitude of ML and deep learning methods show promise. Recurrent neural networks (RNNs) have demonstrated enhanced FoG detection performance by integrating spectral data from neighboring time windows, without requiring a longer analysis window. Innovations like DeepFoG [71] employ real-time FoG detection using a single-arm inertial measurement unit (IMU), facilitating therapeutic interventions such as rhythmic auditory stimulation (RAS) and hand vibration on Physionet data [70]. Supervised ML classifiers, such as decision trees, SVM, ensemble classifiers, and bagging classifiers, have reported high classification accuracy (up to 99.4%) in differentiating PD stages based on temporal and spatial gait features using 10-fold cross-validation [72]. Various FOG detection techniques such as CNN, RNN, and LSTM with different cross-validation approaches were evaluated on a dataset collected in home environments using a single tri-axial sensor, employing both supervised and self-supervised methods with ML and DL algorithms. The results show that combining spectral features

from previous windows with CNN and LSTM layers enhances detection accuracy while maintaining low latency, essential for real-time monitoring and reducing FOG events [73]. The dataset tested in this study was the Daphnet Freezing of Gait dataset [74], created by Tel Aviv Sourasky Medical Center and ETH Zurich, which includes 3D acceleration data from wearable sensors on the ankle, thigh, and hip of 10 PD patients performing tasks to induce gait freezing, with data from 5 subjects used for analysis. The authors proposed a novel framework modeling spectral information of adjacent windows through an RNN that has higher performance accuracy for FOG detection when compared against several ML models using a leave-one-subject-out (LOSO) cross-validation. LSTM and transfer-learning based method were compared for the early detection of freezing of gait (FoG) which is a symptom of PD [75]. In an approach to diagnose cerebral palsy gait [76], a Gaussian radial basis function (GRBF) kernel SVM achieves an 83.33% accuracy in analyzing gait features on CP and normal children's gait data [77]. A detailed overview of the methods, models, performance metrics, and references is provided in Table 5, drawn from studies that focus on PD detection through machine learning and deep learning approaches, utilizing gait and tremor analysis. These works have proved that monitoring changes in gait over time is crucial for tracking PD progression, optimizing treatment plans, and designing individualized rehabilitation strategies.

Table 5. A summary of the state-of-the-art machine and deep learning methods applied on gait dataset for PD diagnosis and FoG detection.

Method	Model	Performance	Reference
Support Vector Machines (SVMs)	Gait Analysis + Tremor	83% accuracy	[68]
SVM + Linear Discriminant Analysis (LDA), ANOVA	Gait Analysis + Tremor, Feature Identification	92.25% accuracy	[67]
Random Forest + Recursive Feature Elimination (RFE), SVM +RFE	Gait Analysis	96.67 accuracy	[69]
LOSOCV, 10-fold CV	Freezing of Gait Detection	83%, 86% sensitivity 88%, 90% specificity	[71]
Decision Trees, SVM, Ensemble classifier, Bagging	PD Stage Classification	Up to 99.4% accuracy	[72]
CNN, LSTM, RNN with Triaxial Accelerometers	Freezing of Gait Detection	Up to AUC ROC of 0.936	[73]
LSTM + Transfer learning	FOG Prediction	87.54% accuracy	[75]
Gaussian Radial Basis Function (GRBF) kernel SVM	Gait Analysis	83.33% accuracy	[76]

Along with important takeaways from the above table, the bullet points below highlight developments in employing ML to diagnose PD using gait.

- Gait analysis aids PD classification by examining temporal (e.g., stride time, cadence) and spatial parameters (e.g., step length), as well as kinetic features like peak vertical forces.
- ML enhances gait analysis by automating feature extraction, improving early PD detection. SVM, random forest, and neural network algorithms are commonly used, with RFE for selecting clinically relevant features. Random forest with RFE attains 96.67% accuracy, and ensemble methods report up to 99.4% accuracy in PD stage classification.
- Tremor analysis, combined with gait metrics, improves diagnosis accuracy, with SVM and LDA reaching up to 92.25%.
- Cross-validation methods like LOSO and 10-fold ensure model robustness, achieving up to 99.4% accuracy for PD staging.

- RNNs and inertial measurement units (IMUs) show promise for detecting freezing of gait (FoG), a critical symptom of PD. For FoG detection, CNN, RNN, and LSTM models using triaxial accelerometer data achieve high accuracy, with AUCs up to 0.936.
- Transfer learning and LSTM predict FoG events with 87.54% accuracy, supporting early intervention.
- Gait analysis also distinguishes similar disorders (e.g., cerebral palsy), with GRBF SVM achieving 83.33% accuracy.

Overall, these findings highlight the significant advancements in gait analysis through ML, emphasizing its role in early PD detection, disease monitoring, and tailored rehabilitation strategies.

3. Current Trends and Limitations

In summary, the current trends in PD diagnosis via machine learning are characterized by the integration of advanced neural networks, hybrid models, and multimodal biomarkers (speech, EEG, fMRI), which are enhancing diagnostic precision and paving the way for more personalized and real-time monitoring solutions which reflects an increasing focus on leveraging advanced computational techniques. The application of machine learning, particularly deep learning models like CNNs, RNNs, and LSTM, has automated speech analysis, leading to higher accuracy in distinguishing PD patients from healthy controls. Current approaches focus on using multiple frameworks and advanced algorithms like SVM and Bidirectional LSTM for capturing dynamic features of speech, achieving impressive classification accuracy (up to 98%). EEG-based diagnosis trends emphasize using advanced neural network architectures like CNNs and hybrid CNN-RNN models. These methods demonstrate notable improvements in classification accuracy and predictive performance (e.g., up to 99.5% accuracy). Real-time monitoring and personalized treatment through wearable EEG devices are emerging areas, driven by data augmentation and transfer learning. Machine learning models, such as DBV and SVM, are used to process neuroimaging data. New trends involve using deep learning for automated feature extraction from fMRI scans, facilitating the early detection of PD. Researchers are also focusing on using functional connectivity mapping to reveal the impact of PD on brain networks, achieving classification accuracies up to 95%.

The current state-of-the-art machine learning (ML) models for Parkinson's disease (PD) diagnosis face several limitations. One major challenge is the limited size and availability of datasets, including those based on EEG, MRI, handwriting, and speech. These datasets often lack the breadth necessary to train generalized ML and deep learning models effectively. Small dataset sizes can lead to overfitting, where models perform well on training data but fail to generalize to new, unseen data, compromising their reliability. Additionally, many of the dataset modalities used in testing ML approaches are not commonly employed in clinical practice. Clinicians primarily rely on tools such as the Unified Parkinson's Disease Rating Scale (UPDRS) and DatScan imaging for diagnosis and staging. This disconnect means that ML models are often trained on data that do not align with standard clinical practices, potentially limiting their applicability in real-world settings.

Another significant limitation is the lack of interpretability in ML models. While these models can classify PD and its stages with high accuracy, understanding and explaining how decisions are made is crucial for clinical adoption. The ability to visualize and interpret features captured by different layers of the models, as well as identify which samples influence the model's predictions, is essential for building trust and ensuring that the tools are useful for clinicians.

Furthermore, the accuracy and availability of annotations from medical experts are vital for training supervised learning models. However, obtaining these annotations can be costly and impractical, particularly for large, diverse datasets. This poses a challenge for creating robust models that can handle a wide range of patient data. Combining predictions from multiple ML methods and data modalities may enhance diagnostic accuracy and sensitivity. Clinicians use various biomarkers to form a diagnosis, and integrating

these different sources of information through advanced ML techniques could offer more nuanced insights.

Most existing datasets include patients with advanced symptoms, which limits the ability to detect early biomarkers of PD. The development of new datasets and tools that can identify early-stage biomarkers is essential. This endeavor requires close collaboration between AI researchers and medical professionals to create and validate new models based on prospective data that could reveal the disease at its onset. Hence, while ML models hold great promise for PD diagnosis, addressing these limitations—such as dataset size, clinical relevance, interpretability, and early detection—will be crucial for advancing these technologies and improving patient outcomes.

4. Future Trends and Discussion: Integration of Biomarkers

The utilization of machine learning in PD diagnostics emphasizes the integration of multiple biomarkers using sophisticated data fusion techniques and federated learning frameworks. By incorporating various modalities such as speech, EEG, fMRI, and gait patterns, researchers aim to develop more comprehensive and accurate diagnostic tools. This integrated approach enhances early PD detection, facilitates personalized treatment plans, and ultimately improves patient outcomes. A wide spectrum of biosignals has been investigated to identify robust biomarkers for PD. For example, EEG signals have been utilized to classify gait patterns, allowing the differentiation between PD patients and healthy controls. This is achieved using advanced signal processing techniques and machine learning algorithms that provide a nuanced analysis of these complex data streams [78]. One notable advancement in this field involves the integration of EEG and MRI data within a multimodal machine learning framework, achieving superior results in early PD detection. This method, employing the Light-GBM algorithm, reached a remarkable 97.17% accuracy, surpassing traditional methods in both precision and insights into early-stage PD [79]. Additionally, another study combining EEG and fMRI, using the AdaBoost algorithm, reported an impressive 93.45% diagnostic accuracy on the openneuro dataset, further validating the potential of such multimodal approaches [80].

Despite the promising results, integrating multiple biosignals—such as speech, EEG, fMRI, and gait—remains challenging due to the temporal variability in the data collection during patient assessments and the need for longitudinal monitoring of disease progression. The interdependence between certain modalities, such as gait and EEG, or EEG and fMRI, remains underexplored, which presents a valuable opportunity to optimize biomarker combinations for enhanced diagnostic accuracy. Current research on multimodal biomarker fusion in automated PD assessments has predominantly employed two key fusion strategies: feature-level fusion (early fusion) and model-level fusion (late fusion). Feature-level fusion involves the extraction and merging of features from multiple biosignals into a unified representation, using methods such as feature concatenation or advanced dimensionality reduction techniques like principal component analysis (PCA) and canonical correlation analysis (CCA). This approach is particularly useful for managing high-dimensional data, but it requires computational efficiency to mitigate the complexity. On the other hand, model-level fusion involves training independent models for each biosignals modality and subsequently combining their outputs through ensemble techniques such as majority voting, weighted averaging, and stacking [81–83]. In the realm of hybrid brain–computer interfaces (BCIs), the combination of EEG with other modalities like near-infrared spectroscopy (NIRS) has shown improved performance. Banville et al. demonstrated that fusing EEG and NIRS data for mental task classification produced better accuracy than using either modality independently. Their approach involved extracting log power from EEG channels and chromophore concentration from NIRS, applying a linear kernel SVM to each modality and their fusion, leading to enhanced classification outcomes [81–83]. Moreover, integrating clinical records with medical imaging has demonstrated considerable promise in medical diagnostics. For instance, one study achieved an area under the receiver operating characteristic curve (AUROC) of 0.947 for automated

pulmonary embolism detection using a late fusion model, while another study utilizing structural MRI and DTI data for Alzheimer’s disease classification via SVM significantly improved diagnostic precision [84,85]. Hashing models have also been explored in multi-modal fusion tasks, providing advantages such as memory efficiency and robustness to data changes, while maintaining semantic similarity in lower-dimensional spaces [86].

Federated learning (FL) represents an innovative solution to the challenges of data integration in PD diagnostics, particularly in the context of protecting patient privacy and addressing the small sample sizes common in medical datasets. Originally developed by Google, FL allows multiple client sites to collaboratively train a global model without sharing sensitive data. A central server initializes the global model, distributing its weights to local clients, who then train models on their private datasets. The server aggregates these updates iteratively, optimizing the global model while maintaining data privacy [87]. “Federated averaging” is a crucial technique that accounts for the size of local datasets in this weight aggregation process, improving the model’s performance in diverse settings [88]. Three main subtypes of FL have emerged: horizontal federated learning (HFL), which operates on datasets with similar features across users; vertical federated learning (VFL), which integrates diverse feature sets across users; and federated transfer learning (FTL), which allows for learning tasks across disparate datasets by leveraging knowledge transfer from related domains [89].

FL has shown rapid growth in medical diagnostics, particularly in rare disease detection, where data scarcity can significantly impact the accuracy of traditional machine learning models [90,91]. By pooling insights from multiple clinical sites, FL enhances sample sizes, increases model generalizability, and reduces the impact of limited datasets. Existing machine learning architectures, including U-Net models, have been successfully adapted to FL frameworks for tasks such as medical image segmentation [92,93]. In the realm of electronic health records (EHRs), FL enables the decentralized training of machine learning models while ensuring strict patient confidentiality. FL also addresses challenges like data heterogeneity and temporal complexity in EHRs, outperforming traditional methods [94]. Further advancements in FL include domain-specific adaptations, such as using an encoder–decoder structure for magnetic resonance (MR) image reconstruction [95,96], as well as applying CNNs and graph neural networks (GNNs) to manage domain shifts in medical imaging [97]. Techniques like ensemble modeling, generative adversarial networks (GANs), and contrastive learning have been introduced to align performance across heterogeneous datasets, demonstrating improved model robustness and accuracy [98–100]. These methods are particularly effective in rare disease analysis, including PD, where data variability and scarcity are significant hurdles. Weight aggregation techniques [101–103] play a crucial role in enhancing FL’s performance. For instance, the Progressive Fourier Aggregation method integrates low-frequency components of model parameters essential for deep learning, while client-specific training loss adjustments, such as those proposed by Li et al., improve the aggregation process in imbalanced datasets, such as in COVID-19 detection [104–106]. FL’s adaptability across medical imaging tasks, such as pancreas segmentation [107], thyroid image classification [108], and anal cancer prediction [109], underscores its versatility in healthcare applications. The application of FL to wearable devices in PD research has shown promising results in improving motor function classification, especially in scenarios with intermittent data transmission. Methods like Fed-Avg and Fed-Prox have demonstrated scalability and performance enhancement, positioning FL as a valuable tool in neurodegenerative disease diagnostics [110,111]. A comprehensive summary of current trends in multimodal biomarker integration and their applications are listed on Table 6.

Integrating biomarkers like speech, EEG, fMRI, and gait data with ML enhances PD diagnostic accuracy, supporting early detection and personalized treatment. Together with important details from the above table, the bullet points below describe developments in multimodal integration techniques for possible PD diagnosis applications.

- Combining EEG and MRI has shown promise, with models like Light-GBM and AdaBoost achieving 97.17% and 93.45% accuracy. Feature-level (early fusion) and model-level (late fusion) fusion are key multimodal integration strategies.
- Federated learning (FL) offers a privacy-preserving solution to data integration challenges by allowing models to be trained collaboratively across multiple sites without sharing sensitive patient data. FL variants, such as horizontal (HFL), vertical (VFL), and federated transfer learning (FTL).
- Recent adaptations of CNNs, GNNs, and U-Net within FL frameworks aid medical diagnostics, particularly for rare diseases and image segmentation.
- Ensemble modeling, GANs, and contrastive learning increase robustness in models for heterogeneous data, benefiting PD and neurodegenerative disease diagnostics.
- Advanced weight aggregation in FL, such as Progressive Fourier Aggregation, enhances model performance on imbalanced datasets, proving effective in healthcare, including COVID-19 and wearable-based PD diagnostics.

Table 6. A summary of recent trends for biomarkers integration for the potential application of PD diagnosis.

Application	Method	Reference
Integration of multiple biomarkers using machine learning	Data fusion techniques	[78]
Combination of EEG, and fMRI	Light-GBM with FL framework	[79]
Combination of EEG and fMRI	AdaBoost with Fusion model	[80]
Integration of clinical records with medical imaging	Late fusion model, FL	[82,84,85,94]
Federated learning for privacy preserving data sharing	Federated learning (FL)	[87]
Federated learning for similar, diverse, and disparate dataset	HFL, VFL, FTL	[89]
Adaptation of existing machine learning architectures to FL frameworks	CNNs, GNNs, UNet	[92,93,96,97]
Integration of heterogenous dataset, improved robustness, and accuracy	Ensemble modeling, GANs, contrastive learning	[98–100]
Weight aggregation techniques for improved FL performance	Progressive Fourier Aggregation	[104–106]

Recent breakthroughs in PD diagnosis using machine learning and FL have paved the way for more precise and individualized care. By integrating diverse biomarkers with fusion models and employing FL for privacy-preserving, distributed learning, researchers are addressing the challenges of small datasets and heterogeneous data distributions, particularly for rare and complex diseases like PD. As FL frameworks continue to evolve, they are likely to play an increasingly critical role in advancing diagnostic accuracy and treatment personalization, facilitating collaborative research across diverse clinical environments.

5. Conclusions

In conclusion, this comprehensive review underscores the critical role of machine learning (ML) in the detection and diagnosis of PD through the analysis of various biomarkers, including speech, EEG, fMRI, and gait patterns, both individually and in combination. The reviewed literature highlights significant advancements achieved with ML algorithms, particularly deep learning models such as CNNs and RNNs, which demonstrate substantial promise for the early diagnosis and ongoing monitoring of PD. ML models, including CNNs, RNNs, LSTMs and hybrid architectures, have demonstrated diagnostic accuracies between 82% and 98% across various modalities such as speech, EEG, fMRI, and gait analysis. This review also discusses how the widely implemented SVMs handle small, structured datasets well but struggle with high-dimensional neuroimaging data, while CNNs and LSTMs excel with complex data but risk overfitting and lack interpretability. The integration of diverse biomarkers with sophisticated ML techniques is anticipated to en-

hance diagnostic accuracy and enable personalized treatment strategies, ultimately leading to improved patient outcomes. However, several challenges persist, such as the necessity for comprehensive datasets encompassing a multitude of relevant biosignals, enhancing model generalizability, and effectively integrating multimodal biomarkers. Future research should focus on addressing these challenges to further propel the field forward. Looking ahead, the future of PD diagnosis through the combination of multiple biosignals will rely heavily on advanced ML techniques to integrate various modalities, including speech, EEG, fMRI, and gait analysis. The literature survey supports the potential adoption of federated learning (FL) frameworks, which can facilitate biomarker integration without requiring a centralized, comprehensive dataset containing heterogeneous data. This approach is expected to significantly improve model generalizability, especially in the context of rare or complex diseases. The implementation of multimodal fusion approaches, particularly when combining biomarkers like fMRI and EEG, shows significant potential in improving diagnostic precision. The use of FL was identified as a future avenue for enhancing collaborative research without compromising data privacy, especially in the case of rare neurodegenerative diseases. Additionally, the application of advanced ML algorithms to merge biosignals promises to enhance the early detection of PD with increased accuracy. The emergence of hybrid models and self-supervised learning techniques further indicates a shift towards developing more precise and personalized diagnostic tools. Collectively, these trends are positioned to advance early detection, optimize treatment strategies, and ultimately improve patient outcomes in PD.

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